JAEHYUN LEE

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EDUCATION

HONGIK UNIVERSITY — Seoul, South Korea
M.S. in Materials Science and Engineering — *Expected Mar 2027*
B.S. in Materials Science and Engineering — *Mar 2025*
Thesis: Strain-Engineered and H_2O_2-Etched MoS_2 for Hydrogen Evolution Reaction

RESEARCH INTERESTS

- Plasma-Assisted Surface Modification
- Spectroscopic Analysis of Surfaces and Interfaces
- Ionomer-Free Electrode Architectures
- Atomic-Scale Simulation of Defects and Interfaces

RESEARCH EXPERIENCE

Graduate Research Assistant — *Mar 2025 – Present*
1Q Lab, Hongik University — Seoul, South Korea
Advisor: Prof. Won-kyu Lee (Link)
- Mechanical exfoliation-based dry transfer of wafer-scale 2D $PtSe_2$ for PEM water electrolyzers to eradicate ionomer poisoning.
- O_2 plasma-induced networked clustering of Pt from $PtSe_2$ for minimal Pt loading.
- Spectroscopic analysis of multi-domain $MoTe_2$/$Ni(OH)_2$, oxidized $NiTe_2$, and CuS.

Undergraduate Research Intern — *Apr 2023 – Feb 2025*
1Q Lab, Hongik University — Seoul, South Korea
- Elastomer-mediated strain engineering of layered transition metal dichalcogenides (TMDs) for ultraefficient hydrogen evolution catalysts.
- Defect engineering of TMDs by chemical etching and thermal annealing.

XPS Instrument Operator & Analyst — *Feb 2025 – Present*
Cooperative Center for Research Facilities — Seoul, South Korea
Supervisor: Prof. Chanyoung Yoo (Link) and Prof. Dongwook Lee (Link)
- Operate XPS and analyze data for internal/external measurement requests.

PUBLICATIONS

(Published)

†: Equal contribution
*: Corresponding author

1. S. K. Han,† Y. Kweon,† H. Jeon,† J. H. Cho, I. Kim, J.-H. Yim, **J. Lee**, J. Heo, S. Shin, S. J. Baek, M. K. Cho, Y. J. Jang, S. Y. Kim, I. S. Kim,* M. Kim,* and W.-K. Lee,* "Amorphized and Oxidized Layered $PtSe_2$ as an All-in-One Electrocatalytic Platform for Hydrazine-Assisted Water Splitting," *Advanced Materials* **2026**, e75207. (Featured on National Research Foundation of Korea News) DOI

2. S. Kang,† J. Pyo,† M. Jung, S. Park, T. Kim, H. Jeon, **J. Lee**, W.-K. Lee, Y.-W. Lee,* S. Cha,* and S. Pak,* "Tunable Bandgap of Copper Sulfide via Cation Exchange as a p-Type Transparent Electrode," *Nanoscale* **2026**, 18, 37, 20335–20346. DOI

3. S. Shin,† I. Cho,† S. K. Han,† J. Heo, J. Han, H. Jeon, **J. Lee**, M. K. Cho, D. J. Preston, I. S. Kim,* B. Shong,* and W.-K. Lee,* "Strain-Engineered Oxygen-Modified Nickel Telluride/Nickel Oxide Heterostructures for Bifunctional Alkaline Water Electrocatalysis," *ACS Nano* **2025**, 19, 51, 42796–42815 (Back Cover article) DOI

4. J. Han,† M. K. Nam,† S. Shin,† I. Cho, H. Jeon, J. Heo, **J. Lee**, M. K. Cho, D. J. Preston, M. C. Hersam, I. S. Kim,* B. Shong,* and W.-K. Lee,* "Efficient and Stable Electrocatalytic Oxygen Evolution from $MoTe_x/Ni(OH)_2$ Heterostructures," *Journal of the American Chemical Society* **2025**, 147, 41096−41108 (Featured on National Research Foundation of Korea News) DOI

5. H. Jeon,† H. J. Kwon,† **J. Lee**,† S. K. Han, H. Kim, J. Heo, J. Han, S. Shin, J. Park, M. K. Cho, D. J. Preston, I. S. Kim,* M. Kim, and W.-K. Lee,* "Strain-Enabled Band Structure Engineering in Layered $PtSe_2$ for Water Electrolysis under Ultralow Overpotential," *ACS Nano* **2025**, 19, 9107-9120 (Featured on Hongik News) DOI

(Under revision) **J. Lee**,† S. Shin,† H. Jeon,† D. Lee, T. Zhou, I. S. Kim,* and W.-K. Lee,* "Beyond Monolayers: Activating Multilayer and Bulk Transition Metal Chalcogenides for Water Electrolysis," *Materials Horizons* **2026**

(In preparation) **J. Lee**, Y. Kang, and W.-K. Lee,* "Networked Pt Clusterization on Mechanically Exfoliated $PtSe_2$ as an Ultra-Low-Pt-Loading Platform for Proton-Exchange Membrane Electrolyzer," **2026**

CONFERENCES

1. **J. Lee**, H. Jeon, S. Han, J. Han, S. Shin, J. Park, and W.-K. Lee, "Strain-Driven Electronic Modulation of Layered $PtSe_2$ for Advanced Hydrogen Evolution Catalysis," 2025 *MRS Fall Meeting & Exhibit*, poster 2025, December, Boston, USA. Link

(Upcoming)

2. **J. Lee**, H. Jeon, S. Han, J. Han, S. Shin, J. Park, and W.-K. Lee, "Ionomer-Free Cathode via Dry Transfer of O_2 Plasma-Activated $PtSe_2$ for PEM Water Electrolysis," 2026 *MRS Fall Meeting & Exhibit*, poster 2026, December, Boston, USA.

SKILLS

SURFACE CHARACTERIZATION

- XPS and UPS
- XRD
- UV-Vis and FT-IR Spectroscopy
- Raman Spectroscopy
  - In Situ Characterization
- SEM, TEM and AFM
- Contact Angle Measurements

MATERIALS TUNING

- Ar, N_2, O_2 Plasma-Assisted Etching
- Elastomer-Utilized Strain Engineering
- Electrochemical Dissolution/Oxidation
- Calcination, Sintering and Annealing
- Tip/Bath Sonication

MATERIALS SYNTHESIS

- Solution-Based Sulfidation
  - Copper Sulfide
- Thermal Evaporation
- RF/DC Sputtering
- Electrodeposition/Plating
- Spin-Coating, Polishing, Ink Making

ELECTROCHEMICAL ANALYSES

- Potentio-/Galvanostatic Operation
- Three-/Two-Electrode Testing
  - Cyclic/Linear Sweep Voltammetry
  - Tafel Analysis
  - Electrochemical Impedance Spectroscopy
  - Redox Characterization
- Gas Chromatography
- In Situ Flow-Cell Characterization

DEVICE FABRICATION

- Membrane Electrode Assembly
- Dry/Wet Catalyst Loading
- 2D Material Mechanical Exfoliation
- Nanoimprinting and Molding

PROGRAMMING & LANGUAGE

- MATLAB, Python, Blender
- Korean and English (Native), Japanese (Intermediate), Chinese (Conversational)

AWARDS

Academic Excellence Scholarship *Spring 2024*
Undergraduate Research Assistantship Scholarship *Mar 2024 – Mar 2025*
National Science & Technology Scholarship *Mar 2025 – Mar 2027*

PROJECTS

Patent Landscape Analysis: Chiplet Packaging *Spring 2023*
Campus Patent Universiade Competition
Advisor: Prof. Kiyoung Lee (Link)

- Analyzed over 3,600 Korean and U.S. patents on 2.5D, 3D, and hybrid chiplet packaging, including Intel and TSMC; identified core patents, technological gaps, and IP/R&D strategies.

Strain-engineered and H_2O_2-etched MoS_2 for Hydrogen Evolution Reaction
Undergraduate Capstone Project *Spring 2024*

- Engineered H_2O_2-induced sulfur vacancies and strain-mediated phase transitions in MoS_2 to build 2H/1T heterointerfaces; achieved HER activity approaching benchmark Pt, validated by XPS and electrochemical characterization.

EXPERIENCES

REPUBLIC OF KOREA/UNITED STATES ARMY Pyeongtaek, South Korea
94th Military Police Battalion, 8th Army, United States Forces Korea
Military Police Officer and Army Interpreter *Apr 2019 – Sep 2021*

- Conducted joint security patrols and law-enforcement operations with U.S. Army Military Police and the Republic of Korea National Police.

OUTREACH

South Africa Vastfontein Program Pretoria, South Africa
Student Mentor *Nov 2017*

- Supported fundraising initiatives for children's daily needs and education.

Community Meal Program Suwon, South Korea
Volunteer *Oct 2020*

- Assisted with meal preparation and distribution for underserved communities.

REFERENCES

Prof. Won-Kyu Lee | Link
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